PARISHRAM 2.0

CLASS 12TH 2026

d & f-Block

CHEMISTRY | LECTURE-3

BY SANYA NAYAR MA'AM



TOPICS TO BE COVERED

1 **Properties Of Transition Elements**




ROAR
Tiger hai tu Tiger! Roar

Let's Recap

When Tiger Ma'am says-

Pehle zara purana revise kar lete hai



Ionisation Enthalpy

$$I.E \propto \frac{1}{size}$$

3d

1st I.E.

↳ Irregularity in size

↳ Irregularity in E.C

Properties of Transition Elements

- In a group,

3d
↓ Size ↑, I.E ↓
4d
↓ I.E ↑ (Lanthanoid contraction: Poor screening of 4f orbital e⁻s)
5d

$3d < 4d \approx 5d$
↳ Due to Lanthanoid contraction

Gp-3
Sc → Y → La
Sc → Y: Size ↑, IE ↓
Y → La: Size ↑, I.E ↓

Hg: highest IE_1 (1007 KJ/mol)

La: Lowest IE_1 (540 KJ/mol)

Question

Out of Mn and Fe, which has higher IE_3 and why?

$_{25}Mn : 3d^5 4s^2 \xrightarrow{-1e^-(IE_1)} Mn^+ : 3d^5 4s^1 \xrightarrow{-1e^-(IE_2)} Mn^{2+} : 3d^5 \xrightarrow{-1e^-(IE_3)}$

Half filled Stable E·C bcoz of which IE_3 becomes high.

$_{26}Fe : 3d^6 4s^2 \longrightarrow Fe^+ : 3d^6 4s^1 \longrightarrow Fe^{2+} : 3d^6 \longrightarrow$



In 3d series which element has the highest IE_3?

Sc^{2+} Ti^{2+} V^{2+} Cr^{2+} Mn^{2+} Fe^{2+} Co^{2+} Ni^{2+} Cu^{2+} Zn^{2+}

At. No. ↑, Size ↓

Zn: Highest IE_3

$Zn^{2+} \rightarrow [Ar]^{18} 3d^{10}$



Question

In 3d, 4d and 5d series which element is expected to have highest IE_1?

$Hg : [Xe]^{54} 4f^{14} 5d^{10} 6s^2$



Why Cr and Cu have abnormally higher 2nd IE?

$_{24}Cr : 3d^5 4s^1 \xrightarrow[-1e^- (IE_1)]{} 3d^5$ (Half filled stable E·C)

$_{29}Cu : 3d^{10} 4s^1 \longrightarrow 3d^{10}$ (Fully filled stable E·C)

Both → Abnormally high IE_2



Oxidation State

d-block elements show variable O.S. (except 1st and last member) due to similar energies of ns and (n – 1)d electrons.

Table 4.3: Oxidation States of the first row Transition Metal (the most common ones are in bold types)

$3d^2 4s^2$ $3d^3 4s^2$

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
	+2	+2	+2	**+2**	**+2**	**+2**	**+2**	+1	**+2**
+3	+3	+3	**+3**	+3	**+3**	**+3**	+3	**+2**	
	+4	+4	+4	+4	+4	+4	+4		
		+5	+5	+5					
			+6	+6	+6				
				+7					

Question



Name the transition element which does not exhibit variable oxidation state.

Ans: - Sc

Calculate the oxidation state of Fe in $Fe(CO)_5$ and Ni in $Ni(CO)_4$.

CO : Neutral

$Fe(CO)_5$

$x + 0 = 0$

$x = 0$

$Ni(CO)_4$

$x + 0 = 0$

$x = 0$

Name any two transition metals which exhibit oxidation state of +8.

Grp 8

Fe → +6

Ru, Os → +8

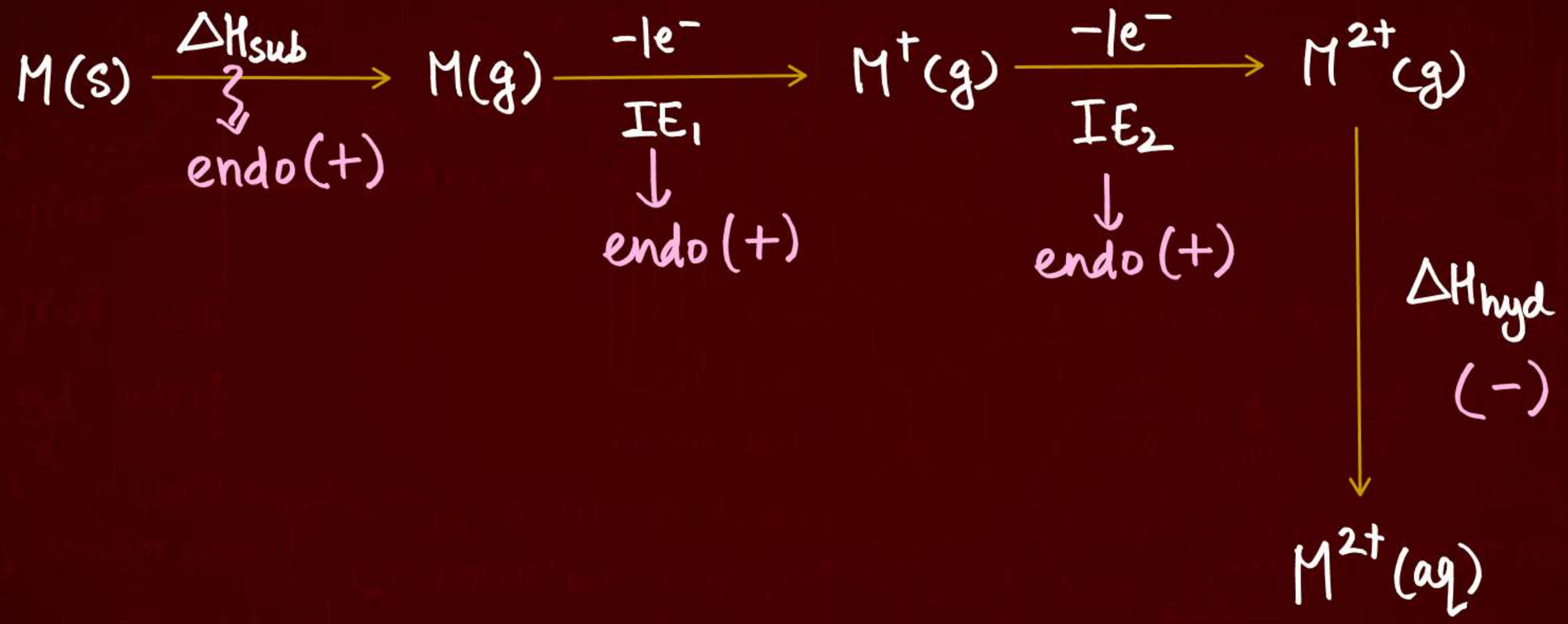
Ruthenium

Osmium

Trends in the M^{2+}/M Standard Electrode Potentials

$E°/V$	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
(M^{2+}/M)	-1.63	-1.18	-0.90	-1.18	-0.44	-0.28	-0.25	0.34	-0.76

$E°(M^{2+}/M)$ values are irregular due to irregular variation of I.E. ($\Delta_iH_1 + \Delta_iH_2$) and also sublimation enthalpies which are relatively much less for Mn & V.


M(s)
ΔHsub
endo(+)
M(g)
-1e⁻
IE1
endo(+)
M⁺(g)
-1e⁻
IE2
endo(+)
M²⁺(g)
ΔHhyd
(-)
M²⁺(aq)

Trends in the M^{2+}/M Standard Electrode

- **Cu : positive E°**

This is because high energy is required to transform Cu(s) to $Cu^{2+}(aq)$ which is not balanced by its hydration energy.

$$Cu(s) \xrightarrow[(+)]{\Delta H_{sub}} Cu(g) \xrightarrow[(+)]{IE_1} Cu^{+}(g)\ (3d^{10}) \xrightarrow[(+)]{IE_2} Cu^{2+}(g)\ (3d^{9}) \xrightarrow[(-)]{\Delta H_{hyd}} Cu^{2+}(aq)$$

Energy reqd > Energy released

Question

In 3d series Mn shows highest oxidation state. Why?



+2 to +7 O.S

Due to similar energies of 3d & 4s electrons

Question



Why is Cr^{2+} reducing and Mn^{3+} oxidizing when both have d^4 configuration?

$Cr^{2+} \rightarrow 3d^4$

(R.A) → loss of e^-

$Cr^{2+} \xrightarrow{-1e^-} Cr^{3+}$

$3d^4 \longrightarrow 3d^3$

1	1	1		

Half filled

Stable t_{2g}^3 E.C

$Mn^{3+} \rightarrow 3d^4$

(O.A) → gain of e^-

$Mn^{3+} \xrightarrow{\text{gain } e^-} Mn^{2+}$

$3d^4 \longrightarrow 3d^5$

Half filled stable E.C

Properties of Transition Elements

1. **Density :** Left → Right; density ↑ses
   e.g. Ti (Z = 22) to Cu(Z = 29) density ↑ses

2. **Metallic bonding:** Greater the no. of unpaired electrons, stronger is the resultant bonding.

Metallic Bonding ∝ No. of unpaired electrons

Metallic Bonding ∝ Hardness ∝ M.P and B.P

Properties of Transition Elements

3 **M.Pt and B.Pt:** High (except Zn, Cd, Hg) due to strong metallic bonds and presence of partially filled d-orbitals.

Gp 12

Zn	$3d^{10}4s^2$
Cd	$4d^{10}5s^2$
Hg	$5d^{10}6s^2$

→ no unpaired e^-

MB Weak, MP↓, BP↓

4. **Enthalpy of Atomisation :** High

It first ↑ses (upto middle) and then ↓ses in a series.

No. of unpaired e^-s ∝ enthalpy of Atomisation

Properties of Transition Elements

5. Magnetic Properties

- **Diamagnetic :** no unpaired e^- (repelled by magnet)
- **Paramagnetic :** One or more unpaired e^- (weakly attracted by magnet)
- **Ferromagnetic :** strongly attracted by magnet

$$\mu_s = \sqrt{n(n+2)} \text{ BM}$$

Spin only magnetic Moment

Bohr Magneton

n = no. of unpaired e⁻s.

Question



Calculate spin only magnetic moment of given ion:

$3d^{10}4s^{1}$

i) $Cu^{2+} \rightarrow 3d^{9}$



$n=1$

$\mu_s = \sqrt{n(n+2)} = \sqrt{1(3)} = \sqrt{3} = 1.732\ BM$

ii) Cr^{2+}

iii) Fe^{3+}

iv) Ni^{2+}

Properties of Transition Elements

6. Complex Formation

- Due to smaller size, high ionic charges on transition metals and availability of d-orbitals for bond formation

Example - $[Fe(CN)_6]^{3-}$, $[Mn(H_2O)_4]^{2+}$ etc.

7. Catalytic Properties

- Due to their ability to adopt multiple oxidation state and ability to form complexes.

Fe	: Haber process $N_2 + 3H_2 \xrightarrow{Fe(s)} 2NH_3$
Ni	: Hydrogenation
MnO_2	: $KClO_3 \rightarrow KCl + O_2$

MnO_2

Question

Why do the transition elements exhibit higher enthalpies of atomization?

↳ T.E have the presence of unpaired e^-s.

Question

In the series Sc (Z = 21) to Zn (Z = 30), the enthalpy of atomization of zinc is the lowest, i.e., 126 kJ mol^{-1}. Why?

$Zn: 3d^{10} 4s^2$

↳ No unpaired e^-s

Low EOA

Question

Which of the 3d series of the transition metals exhibits the largest number of oxidation states and why?

Mn +2 to +7

Due to the similar energies of 3d & 4s electrons

Question

The E° (M^{2+}/M) value for copper is positive (+0.34V). What is possible reason for this?



Homework

→ DPP Solve

→ NCERT Q's

THANK YOU

